

The AI Workshop

A construction kit in a browser where children build **pipelines that learn**: real data of any kind goes in — photographs, poses, spreadsheets — rides a conveyor as crates, is read by real learning algorithms, and comes out as action: a gate thrown, a lamp lit, a number on a display. Then the child finds out, honestly, how much the machine actually understood. **The subject is not how to use AI. It is how to tell whether AI is any good.** This brief is for you to plan lessons against while the product is still being built: what it is, what works today, what is landing, and what is direction only.

DNA	In ethos: Scratch (plain parts, live feedback, nothing locked — the complexity is the student's, never the tool's) and micro:bit (real sensors, machines that act on the world). In shape: a dataflow instrument , closer to Node-RED or LabVIEW than to Scratch — because ML genuinely is a dataflow discipline, and staged pipelines are how the field itself now builds, from model-serving to image tools to agent frameworks. The workshop is that shape at child scale.
Not “Scratch for AI”	Scratch teaches control : you write the logic, the sprite obeys, and a bug is your thinking made visible. The workshop teaches data : the child never writes the rule — they arrange a pipeline, and the machine fits the rule from examples. What transfers is twofold: the practitioner's discipline (split your data, hold some back, train, test on what it has never seen, read the gap) and architectural fluency — decomposing a job into stages, wiring explicit connections, putting monitoring on the result, debugging by tracing. Nobody ships a bare model; someone architects what surrounds it.
What we build	The thing that turns any model into a playable part. Each model gets a face (it draws its actual thinking — the neighbours that voted, the fitted line with its residuals), dials (its real hyperparameters), and a body (it stands on a belt and does real work in a child's machine — taught, tested, fooled, debugged). Models are employed here, not watched.
Who	Upper primary and above, teacher-led. A child drives it; a teacher gives it direction. (Age banding is asserted, not yet classroom-validated — treat it as a working assumption.)
Runs on	Tablet or laptop browser. No install, no accounts, no network — the vision models ship with the app; nothing about a child leaves the device. Live build: workshop.ai-education.workers.dev .
The models	A growing library — six today: k-Nearest Neighbours, Nearest Centroid, k-Means, k-NN regression, ridge regression with polynomial capacity, a small neural network trained live. Every algorithm is a plug-in part, so the shelf grows without the product changing. MobileNet embeddings for vision, MediaPipe for pose.

The twenty parts — the vocabulary you are planning with

Every activity is some arrangement of these. Nothing is locked or earned; the shelf is the whole set. Three plug shapes tell a child what fits what: **square couples belts**, **round takes cables**, **diamond is a dial socket** — nothing invisible ever connects two parts.

Moving things	Feeder (turns whatever arrives into crates on the belt) · Track (a belt, with a speed) · Sorter gate (fans out 2–6 ways; something must throw it) · Buffer (holds crates until released) · Bin (end of a route; counts arrivals).
Real data in	Camera (webcam photo down a cable) · Files (eats a spreadsheet or photo folder, deals it out row by row) · Splitter (deals a table into study / check / sealed-test piles before anything touches it — held-

out data as a physical fact).

Thinking

Model (reads the crate in front of it and answers — or says *not sure*; carries its own sense, algorithm and dials) · **Labeller** (files what passes onto a named Model's shelf — how a machine teaches a machine) · **Evaluator** (opens the crate, compares guess to truth, keeps the two scores apart and draws the picture).

Logic, timing, output

Filter (passes a matching signal) · **Counter** (fires at a number you choose) · **Timer** (fires on a clock, or on silence) · **Window** (sliding average of recent numbers) · **Chance** (a weightable dice) · **Button** (a thing to press) · **Lamp** · **Sound** · **Display** (light up, beep, show a word or number).

The rules that hold everywhere

These are invariants — enforced by the product's test suite; they stay true whatever you design:

- **Evaluation is the spine.** Data is split before anything touches it; the Evaluator always keeps two scores — studied vs. never-met — and the gap between them is overfitting, met as an experience before the word arrives.
- **Nothing invisible ever connects two parts.** Every relationship is a drawn cable, a coupling, or a part standing on a belt. A child who traces the connections has genuinely understood the machine — the pedagogy, made structural.
- **The product cannot lie about the ML.** A model's face is pinned to the real math; any answer's Evidence traces back to the stored examples that caused it; Run resets belt-fed learning so a replay can't fake it.
- **“Not sure” is built in.** Below a confidence threshold the child sets, the machine abstains instead of guessing.
- **Real vocabulary, plain sentence underneath.** k-Nearest Neighbours, Regularisation, Evaluator, confusion matrix — the field's own terms at age ten, nothing to unlearn at fifteen.
- **An in-app AI coach can read any machine.** A child's build is self-describing data; the coach explains it back or changes it, every action visible on the floor. (The one optional feature that uses a network.)
- **Offline, private, portable.** Runs with the network unplugged; photographs stay in memory; a child's whole workshop saves to one file they own.

What a class can do today

Not lessons — the raw experiences the product delivers now, and the shipped machines that deliver them. What to make of them in a classroom hour is yours.

Teach a sorter with their own hands

A camera photographs what the child holds up and drops it on the belt as a crate; the Model reads it *as it rides past* and throws the gate. Taught with six photos from one desk, it fails on another desk — and the failure is theirs to diagnose. Shown something it was never taught, it says *not sure*, and the child owns the dial that decides when.

Shipped machines: rubbish sorter, three-way sorter, watchdog, pose studio.

Predict a number and argue with the rule

A spreadsheet of ice-cream days arrives; a Splitter deals it into three piles — study, check, sealed test — before anything touches it. A button releases the study pile, and the fitted rule comes out in words — “*13 more cups when it's a weekend, 2.8 more per degree*” — filling in on the model's face as it studies. In the data lab, the machine is then examined on rows it held back, and the two scores open the overfitting gap live as a child turns a dial.

Shipped machines: ice-cream stand (study half), data lab (the full examination).

Build systems, not demos

A model's verdict — and separately its *refusal* — are just signals, so children wire what no demo tool allows: two algorithms disagreeing on one belt (an *ensemble*); a specialist that only sees what the first model was unsure about (a *triage cascade*, how production systems actually deploy); a machine watching its own accuracy on a display (*monitoring*); a machine fed its own guesses (a *feedback loop*), which drifts — **model collapse, on a desk, in about a minute**. The named patterns of AI systems architecture, built by hand and debugged by tracing cables.

Shipped machines: clustering sorter, sliding-average watcher, self-training sorter — plus free-form wiring.

The outcomes to design toward

Judgement, not software skills — each learned as a physical experience on the child's own machine:

- **“You cannot test on what you trained on.”** Those rows were held back, so the score means something.
- **Memorising is not learning.** The studied/new gap opens as they turn a dial.
- **“I don't know” is a valid answer.** Their machine abstains at a threshold they chose.
- **A metric is a choice.** Move the tolerance dial and the same machine reads excellent or hopeless.
- **Rubbish in, rubbish out — demonstrated.** Six photos from one desk fail on another desk.
- **Machines that learn from themselves drift.** Model collapse, built on their own desk.

It travels because the loop is the practitioner's own — fit, evaluate, diagnose, adjust — and the words are the field's own. The child who leaves this room asks the only question that matters about any AI: **“how do you know it works?”**

Where this is aimed — any data in, anything out

Scratch and LEGO became generational because one plain grammar could build whatever a child thought of next. That is the ambition here, and the target shape is not ours alone — it is the frontier's. The serving architecture for omni-modality models (**vLLM-Omni**, late 2025) is a *disaggregated pipeline*: encoders, models and decoders as separate stages composed into a graph — any modality in, heterogeneous outputs out, including *actions*. The workshop is already structurally that: stages as parts, one data plane between them, machines whose outputs act. The input half is nearly complete — image, pose, spreadsheet, text, number. The output half still speaks only in labels, numbers, lights and beeps. Closing it needs no new grammar, only new parts, each obeying the same laws (speaks crate/signal; nothing invisible; ships with a worked example):

- **Speaker** — a label becomes speech (on-device TTS already in our toolbox). A machine that talks.
- **Microphone** — speech becomes a word signal. With Speaker: voice in → model → voice out, from parts, on a desk.
- **Answer-with-the-nearest-example** — a model replies with the closest example it holds: image→image, honestly, zero new ML — and the truth about what “generation” often is underneath.
- **The vector made visible** — every modality already becomes numbers in one shared space inside the crate; show it.
- **Tournaments** — one sealed test set for the whole class, best held-out score wins; the Evaluator is already the referee. (Points, skins and unlocks stay banned.)
- **Make-your-own-part** — collapse a working sub-machine into one reusable part with its own ports; Scratch's custom blocks and remix culture sit on this horizon.

For your planning this means the curriculum has a second arc waiting behind the first: today's arc is *judging* a machine (teach, split, test, read the gap); the coming arc is *composing* machines whose inputs and outputs span modalities — the same twenty-part grammar, richer voices. Design the first arc so the second has somewhere to stand.

The planning horizon

What to plan on now, what to plan *around*, and what to only keep in mind. The one gap this brief cannot close is the one it is asking you to write: the product is deliberately an instrument, not a course — lesson spine, pacing and assessment are yours.

PLAN ON IT	Everything above: the full floor (20 parts), the six-model library, camera/pose/spreadsheet input, the Splitter, readable rules, the Evaluator's two scores, the worked example machines, save/load, the live URL. Stable and test-enforced.
DAYS AWAY	The “three acts of Run”: the Model becomes act-driven, which completes the ice-cream stand's own examination half (study, then check, then break the seal on the test pile — on one machine). In development right now; expect the live URL to keep moving.
DESIGNED, NOT BUILT	Two model faces still missing: clustering's piles and the neural network's falling loss curve. Don't build a lesson that needs those two pictures yet.
DIRECTION ONLY	The output half above — Speaker, Microphone, nearest-example answers, the visible vector, tournaments, make-your-own-part. It shapes the curriculum's second arc; nothing in it has a date. Keep in mind; do not schedule.

Companion reading: *The AI Workshop — a briefing for educators* (long-form) covers the pedagogy and practicalities in full; the *block manual* documents every part's plate, ports and dials one by one. · Draft for internal discussion — nothing here is public copy.